

Mingyue Cui

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Research interests: AI safety, LLM interpretability, secure generative models, and embodied AI

Education

The Hong Kong Polytechnic University

Sep. 2023 – Jun. 2027

B.Sc. in Computer Science; Minor in Finance

Hong Kong

- **CGPA:** 3.77/4.3 | **Honour:** Dean's List
- **Core coursework:** Machine Learning, Artificial Intelligence, Big Data Analysis, Operating Systems, Computer Networking, Computer Organization, Database Security, Software Engineering, Linear Algebra, and Discrete Mathematics.

Selected Publications

- **Mingyue Cui**, Zheyuan Liu, and Meng Jiang. "Learning from Runtime Feedback Through Failure-Bank Self-Evolution for Vision-Language-Action Models." *Under review*. [arXiv:2609.39820](https://arxiv.org/abs/2609.39820)
- **Mingyue Cui**, Linghui Shen, and Xingyi Yang. "SAE Interventions are Unreliable: Post-Intervention Recovery of Suppressed Behavior." *Under review*. [arXiv:2606.18322](https://arxiv.org/abs/2606.18322)
- Linghui Shen, **Mingyue Cui**, and Xingyi Yang. "BadWorld: Adversarial Attacks on World Models." *Under review*. [arXiv:2606.16519](https://arxiv.org/abs/2606.16519)
- Linghui Shen, **Mingyue Cui**, and Xingyi Yang. "DeContext as Defense: Safe Image Editing in Diffusion Transformers." *Under review*. [arXiv:2512.16625](https://arxiv.org/abs/2512.16625)
- Chang Liu, Yuni Lai, **Mingyue Cui**, Cong Tian, Yunyan Zhang, Xian Wu, Kai Zhou, and Bin Xiao. "DSPrompt: Dynamic Soft Prompt Defense Against M-RAG Corruption." *Under review*. [arXiv:2608.16536](https://arxiv.org/abs/2608.16536)

Research Experience

University of Notre Dame

Jun. 2026 – Sep. 2026

Undergraduate Researcher, Summer Research / DM2 Group

Notre Dame, IN, USA

Advisors: Prof. Meng Jiang and Zheyuan (Frank) Liu

- Developed **FAILBANK**, a self-evolving VLA safety framework that converts runtime safety-shield corrections into persistent policy improvements via guarded LoRA updates (**PyTorch**, **LoRA**, **VLA**, **CBF**).
- Improved mean success rate from **66.0% to 74.5%** on $\pi_{0.5}$ and **47.5% to 54.4%** on π_0 , while reducing policy-induced safety cost by **35.6%** and **23.8%** across VLA-Arena tasks and two VLA backbones.

The Hong Kong Polytechnic University

Sep. 2025 – Present

Research Assistant, AI Safety and Model Interpretability

Hong Kong

Advisor: Prof. Xingyi Yang

- Led implementation and evaluation of **Sparse Autoencoder (SAE) post-intervention recovery**, building constrained recovery experiments across TPP, WMDP-Bio, IOI, and refusal benchmarks while preserving intervention constraints (**PyTorch**, **TransformerLens**, **SAELens**, **SAEBench**).
- Co-developed **BadWorld**; implemented self-supervised velocity attacks and trajectory-adaptive bi-level optimization for autoregressive world models, and evaluated robustness across varying action sequences.
- Co-developed **DeContext**; implemented and benchmarked image-protection attacks on FLUX.1-Kontext and Step1X-Edit, evaluating perceptual quality, identity similarity, and face quality.

The Hong Kong Polytechnic University

Oct. 2025 – Present

Research Assistant, LLM and Multimodal RAG Security

Hong Kong

STiL Group; Advisors: Prof. Bin Xiao, Kai Zhou, and Dr. Yuni Lai

- Developed a hash-based **multimodal retrieval-augmented generation (M-RAG) defense** that disperses and detects poisoned evidence through multi-query semantic consistency; benchmarked against ReliabilityRAG-MIS on PoisonedEye (**PyTorch**, **FAISS**, **SigLIP**, **LLaVA**).

Industry Experience

ByteDance

Dec. 2025 – Feb. 2026

Machine Learning Intern

Remote

- Built machine-learning and data-analysis pipelines for large-scale content recommendation, covering exploratory analysis, feature diagnostics, and experiment evaluation (**Python**, **SQL**).
- Diagnosed recommendation bottlenecks through user-behavior and feature analysis, informing targeted feature and ranking-model iterations that improved offline **GAUC by 0.3%**.

PolyU School of Accounting and Finance**Jan. 2025 – Aug. 2025***Student Research Assistant**Hong Kong*

- Evaluated LLM-generated educational content, integrated AI tools into digital-business courses, and deployed an interactive learning site (**Jekyll, GitHub Pages**).

Alphabag Limited**Jun. 2025 – Aug. 2025***AI Software Engineer Intern**Hong Kong*

- Led full-stack development of a cross-platform mobile app and integrated an LLM-based letter-generation workflow with PDF/Word export for 200+ parents (**Flutter, Firebase**).

Latent Links Limited**Jun. 2024 – Sep. 2024***AI and Data Marketing Analyst**Hong Kong*

- Conducted competitor analysis, audience segmentation, and data-informed content campaigns using structured data and generative-AI workflows (**Python, SQL, Generative AI**).

Selected Technical Projects

DreamerV3 Reproduction and Policy Optimization**Feb. 2025 – Apr. 2025**

- Reproduced the DreamerV3 model-based reinforcement-learning pipeline on Atari tasks and explored policy-level modifications to improve training efficiency.

Human-Computer Interaction: ClassIn Platform Redesign [Figma]**Sep. 2025 – Nov. 2025**

- Redesigned key ClassIn interaction flows using HCI principles and developed high-fidelity prototypes with usability-oriented interface and workflow improvements (**Figma**).

Software Engineering: Jungle Chess Game [GitHub]**Sep. 2025 – Nov. 2025**

- Developed a command-line Jungle Chess system with modular board-state, rule-engine, and player-interaction components, applying object-oriented software design and collaborative development practices (**Python, Java**).

Multi-threaded HTTP Server**Feb. 2025 – Apr. 2025**

- Implemented a multi-threaded HTTP server with concurrent request handling and optimized networking modules for high-throughput, low-error operation (**Python, Socket Programming**).

Operating Systems: Smart Parking Scheduling Platform**Jan. 2025 – Apr. 2025**

- Designed a dynamic priority-scheduling strategy based on resource utilization and time-window constraints to improve resource-allocation efficiency.

Student Information Database Security [GitHub]**Sep. 2025 – Nov. 2025**

- Implemented layered access control, key management, field-level encryption, and secure-query mechanisms for a student-information management system (**MySQL, SQL**).

Awards

HKSAR Government Scholarship Fund – Talent Development Scholarship (HK\$10,000)	2026
Honorable Mention, Mathematical Contest in Modeling (MCM)	2026
HKSAR Government Scholarship Fund Award (HK\$80,000)	2025
Dean's Honours List, The Hong Kong Polytechnic University	24/25/26
Seoul National University Exchange Nomination (two university-wide places)	2025
Wong Tit-shing Student Exchange Scholarship Nomination	2025
First Prize, 6th College Students' Financial Literacy Competition	2025
Poster: "Optimizing Sports Facility Management Using OS-Level Process Scheduling: A Smart Parking System for Sports Venues," 1st WCSST	2025

Technical Skills

Programming	Python, C/C++, Java, SQL/MySQL, MATLAB, HTML/CSS
ML and research	PyTorch, Hugging Face, LLM/SAE analysis, diffusion models, reinforcement learning, VLA evaluation, adversarial and safety evaluation
Tools	Linux, Git, SLURM/HPC, LaTeX, Jekyll, Flutter, Firebase
Languages	Mandarin Chinese (native); English (professional and academic proficiency)